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DETAIL SPECIFICATION

ROPE, NYLON, EIGHT-STRAND PLAITED

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements of eight-strand plaited nylon rope.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR QUALITY (ASQ)

ANSI/ASQ Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at www.asq.org.)

CORDAGE INSTITUTE

CI-1500A - Test Methods for Fiber Rope Physical Properties

CI-1500B - Test Methods for Fiber Rope Performance Properties

(Copies of these documents are available online at www.ropecord.com.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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3.2 Application. The requirements specified in 3.6 and 3.7 apply only to plaited nylon rope purchased directly by the Government. All other requirements apply to plaited nylon rope purchased as a component for an end item and to plaited nylon rope purchased directly by the Government.

3.3 Materials. The rope shall be fabricated from bright white, virgin continuous filament, heat- and light-resistant nylon fiber of at least 6-denier chain polymer made of hexamethylene diamine and adipic acid, or a long chain polymer of epsilon amino caproic acid. Mixtures of nylon fiber types shall not be employed in any one rope.

3.4 Construction. The rope shall be made of eight strands arranged in four pairs, where one individual strand shall be laid adjacent to the second strand in each pair, and shall conform to the requirements specified herein. Individual strands of ¾- through 4½-inch circumference shall be made of one size of 1-ply yarns; 5-inch circumference and larger shall be made of one size of balanced 3-ply yarns and shall have an equal number of yarns. The single yarns shall be made from group filaments conforming to the sizes specified in table I for the respective sizes of rope.

TABLE I. Construction.

Rope size (circumference) (inches)	Turns per foot, minimum	Denier of single yarns
¾ to 4½	11	5,000 to 35,000
5 to 9	11	15,000 to 35,000
10 to 16	11	24,000 minimum

3.4.1 Preparation. For single yarn construction, the pattern shall be “ZS” and “SZ”. In the preparation of the constituent rope strands, the “SZS” pattern shall be employed in the twisting of the single yarns, the 3-ply yarns, and the individual strands of two pairs, while the components of the remaining pairs shall be twisted in the “ZSZ” pattern.

3.4.2 Finished rope. The finished rope shall be constructed so that in the interweaving procedure, pairs of strands of the former structure shall be twisted in the “Z” direction, while alternating pairs of the latter structure shall be twisted simultaneously in the “S” direction. Heat setting of the rope or any of its components, with the exception of fibers, shall not be permitted.

3.5 Physical requirements. The finished rope shall conform to the physical properties specified in table II, when tested as specified in 4.5. The circumference in accordance with table II shall be as specified (see 6.2). No extraneous material shall be added for the purpose of weighting the rope.

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TABLE II. Physical properties.

Circumference at load P nominal (inches)	Tolerance (±inches)	Diameter nominal (inches)	Load P (200×D²) (pounds)	Linear density +5% or -0% (pounds/100 feet)	Breaking strength, minimum (pounds)
¾	1/16	¼	12	1.35	1,500
1	1/16	5/16	20	2.3	2,500
1⅛	1/16	⅜	28	3.5	3,700
1¼	1/16	7/16	38	4.9	5,000
1½	1/16	½	50	6.2	6,400
1¾	1/16	9/16	63	7.5	8,000
2	⅛	⅝	78	10.2	11,000
2¼	⅛	¾	112	14.1	17,000
2½	⅛	13/16	132	16.2	20,000
2¾	⅛	⅞	153	19.5	24,000
3	3/16	1	200	25.0	31,000
3½	3/16	1⅛	253	33.9	38,000
3¾	3/16	1¼	312	40.0	46,000
4	3/16	1 5/16	345	43.0	53,000
4½	¼	1½	450	50.0	63,000
5	¼	1 ⅝	528	68.0	73,000
5½	¼	1¾	613	82.0	78,000
6	5/16	2	800	95.0	95,000
6½	5/16	2⅛	903	109.0	106,000
7	⅜	2¼	1012	125.0	125,000
7½	⅜	2½	1250	141.0	137,000
8	7/16	2 ⅝	1378	167.0	165,000
9	7/16	3	1800	214.0	200,000
10	½	3¼	2112	262.0	250,000
11	9/16	3½	2450	319.0	300,000
12	⅝	4	3200	384.0	360,000
13	¾	4¼	3612	446.0	380,000
14	¾	4½	4050	517.0	441,000
15	¾	5	5000	595.0	507,000
16	¾	5¼	5512	676.0	572,000

3.5.1 Elongation. The elongation of the ropes shall not exceed 65 percent at the breaking strength (see 4.5.4). The load elongation curve, drawn autographically, shall not exhibit evidence of sudden changes in load applications greater than 5 percent of the load weighed at the instant of change. Changes due to splice slippage shall not be considered in this determination.

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3.6 Identification marker and tag. Products shall be identified by inserting a kraft paper or water repellent cotton marker within one strand in all ropes 2 inches in circumference and larger. The marker shall be completely enveloped by the cover yarns in the strand in which enclosed. Unless otherwise specified (see 6.2), the manufacturer's name, year of manufacture, and type of fiber (nylon) shall be clearly printed on the marker. Italic or script type shall not be used. The printing shall not be affected upon exposure to water or mineral oil. This marker shall extend the entire length of the rope. Additionally, each package unit shall have a tag attached to it for identification purposes. The tag shall contain the following information:

a. Stock number.
b. Nomenclature.
c. Specification number.
d. Length.
e. Contract number and date.
f. Date of manufacture (month and year).
g. Manufacturer's name.

3.7 Put-up. Unless otherwise specified (see 6.2), the rope shall be furnished without knots or splices on non-returnable reels (spools) not larger than 7 feet in diameter. Unless otherwise specified (see 6.2), multiple lengths shall be permitted in the same put-up. When specified (see 6.2), continuous lengths other than a standard length may be used. The rope shall be ordered by weight as specified (see 6.2). Delivered weight per reel shall be within +10/-0 percent of the weight specified in table III. The weight for lengths other than standard shall be determined in accordance with table III, with the proper length ratio applied. However, the actual length shall be not less than the standard length, as specified in table III, when measured in the relaxed condition.

TABLE III. Put-up.

Circumference (inches)	Minimum length (feet)	Weight (pounds)
¾	1,200	18
1	1,200	30
1⅛	1,200	47
1¼	1,200	65
1½	1,200	82
1¾	12,000	102
2	1,200	135
2¼	1,200	187
2½	1,200	214
2¾	1,200	265
3	1,200	332
3½	1,200	448
3¾	600	265
4	600	286
4½	600	333

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TABLE III. Put-up – Continued.

Circumference (inches)	Minimum length (feet)	Weight (pounds)
5	600	49
5½	600	543
6	600	629
6½	600	721
7	600	826
7½	600	932
8	600	1,106
9	600	1,415
10	600	1,734
11	600	2,111
12	600	2,540
13	600	2,951
14	600	3,420
15	600	3,937
16	600	4,472

3.8 Workmanship. The ends of all rope shall be cut off squarely and be securely whipped, taped, or heat sealed. The reels shall conform to the quality and grade of product specified herein.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall be performed on rope when a first article sample is required (see 3.1). This inspection shall include the examinations of 4.4, and the tests of 4.5. The first article sample for ropes shall be a continuous finished length of 70 feet.

4.3 Conformance inspection. A random sample of reels shall be selected from each lot offered to the Government in accordance with ANSI/ASQ Z1.4 at inspection level II and the Acceptance Quality Limit (AQL) specified (see 6.2).

4.3.1 Material and component inspection. Characteristics in table IV shall be verified. Linear density (pounds per 100 feet) shall be determined in accordance with CI-1500A.

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TABLE IV. Component inspection.

Characteristics	Requirement	Inspection method
Material	3.3	Visual
Denier per filament	3.3	Visual
Tenacity (grams per denier)	3.3	Visual
Denier (singles yarn)	3.4	Visual
Turns per foot (singles yarn)	3.4	Visual
Direction of twist (singles yarn)	3.4	Visual
Direction of twist (plied yarn)	3.4	Visual
Direction of twist (strand)	3.4	Visual

4.4 Examinations. A sample shall be examined for the defects specified in table V. If any defects are noted in the sample, the entire lot shall be rejected.

TABLE V. Visual examination and classification of defects

Defect	
Appearance and workmanship	Cut, chafed, and damaged, affecting serviceability
	Kinks, darting yarns [1/] with broken or loose ends, bulged strands, strand knots
	Other than eight strands (four pairs)
	Ends not securely whipped, taped, or heat sealed to prevent fraying or untwisting
Identification marker [2/]	Omitted, incorrect, illegible
	Italic or script type used; not completely covered by cover yarns
	Not as specified
Identification tag	Omitted, incorrect, illegible, or insecurely attached
	Not as specified (hand written entries)
Cleanliness	Spot or stain clearly visible [3/]
	Objectionable color

NOTES:

1/ Darting yarns are internal yarns that project through the cover yarns of the strand at intervals along the rope.

2/ Identification marker for 2 inches in circumference and larger.

3/ At normal inspection distance (approximately 3 feet).

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4.4.1 Defects with regard to length. Defects with regard to length shall be considered to exist if any of the following are determined during inspection:

a. Length of unit less than specified (see 3.7).

b. Length of unit less than marked on tag.

c. Any piece less than 600 feet in length for sizes up to and including ropes of 16-inch circumference.

4.4.2 Defects with regard to winding. Defects with regard to winding shall be considered to exist if any of the following are determined during inspection:

a. Rope improperly or not firmly wound, resulting in kinks, knots, entangling, or slippage during unwinding or otherwise affecting free, unhampered unwinding of rope.

b. Rope ends knotted or spliced to make a continuous length.

4.5 Testing of the end item. Testing of the end item shall be as specified in table VI and 4.5.1 through 4.5.4.

TABLE VI. Test methods.

Characteristics	Requirement	Test method	Number of determinations per individual unit of product	Results reported as
Plied yarn; per strand	3.4	Visual	1	Pass or fail
Direction of rope twist	3.4	4.3.1	1	Pass or fail
Circumference	3.5	4.5.1	3	Pass or fail
Breaking strength of ropes	3.5	4.5.3	3	Pass or fail
Linear density	3.5	4.5.2	1	Pass or fail
Elongation	3.5.1	4.5.4	3	Pass or fail

4.5.1 Determination of circumference. The circumference measurement shall be made in accordance with CI-1500A. Results shall be the average of three determinations to the nearest 1/16 inch.

4.5.2 Linear density. Linear density (pounds per 100 feet) shall be determined in accordance with CI-1500A. Linear density results shall be as follows:

Circumference (inches)	Degree of accuracy (pounds per 100 feet)
Less than 1¾	Nearest 0.01
2 to 6	Nearest 0.10
6½ to 12	Nearest 1.0
More than 12	Nearest 10.0

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4.5.3 Breaking strength. Breaking strength shall be determined in accordance with CI-1500B. Results shall be the average to an accuracy of 1 percent. The load applied need not cause a failure, provided that all individual tests exceed the specified minimum breaking strength in table II.

4.5.4 Elongation. Elongation shall be determined in accordance with CI-1500B.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The rope covered by this specification is intended for shipboard use where relative freedom from torque deformation and high reliability is required, in addition to high strength and high elongation not typically available in commercial products. Current stock of rope may be used until depleted; further requisitions for rope should refer to the current version of this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. When first article inspection is required (see 3.1).

c. Circumference size and weight of rope required (see 3.5 and 3.7).

d. When information on the identification marker is other than specified (see 3.6).

e. Put-up requirements and length, when other than specified (see 3.7).

f. When multiple lengths per unit are not allowed (see 3.7).

g. AQL of 1.0 (see 4.3).

h. Packaging requirements (see 5.1).

i. Acceptance, at original weight, of any unit that has been shortened or cut for test specimen, if in complete compliance with this specification. The reel must be marked as to actual length.

j. That plaited nylon rope will be purchased on a price-per-pound-basis net weight.

6.3 Subject term (key word) listing.

Splices

Strands, rope

Yarns

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Custodian:
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Preparing activity:
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Review activity:
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